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MILITARY SPECIFICATION

LOUDSPEAKER, PERMANENT MAGNET LS-211/AIC-13

This specification is approved for use by all Departments and Agencies of the Department of Defense

1 SCOPE

1.1 Scope This specification covers one type of loudspeaker, designated loudspeaker, permanent magnet, LS-211/AIC-13. This loudspeaker is a horn type unit with a frequency response of 200-7000 Hz, nominal 16 ohm voice coil impedance, and a 20 watt peak power output (see 6.1)

2 APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

SPECIFICATIONS

FEDERAL

GG-C-560	-	Compass, Magnetic, Surveyor's Ball Joint and Jacob's Staff Transit, Pocket
NN-P-71	-	Pallets, Material Handling, Wood, Double Faced, Stringer Construction
QQ-A-250/2	-	Aluminum, Alloy 3003, Plate and Sheet
QQ-S-781	-	Strapping, Steel, Flat and Seals
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-601	-	Boxes, Wood, Cleater-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-T-60	-	Tape Packaging, Waterproof
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive, Paper (For Carton Sealing)

MILITARY

MIL-P-116	-	Preservation-Packaging, Methods of
MIL-C-45662	-	Calibration System Requirements

STANDARDS

FEDERAL

FED-STD-595	-	Colors.

FSC 5965

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-147	-	Palletized Unit loads on 40" x 48" Pallets
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-810	-	Environmental Test Methods

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3 REQUIREMENTS

3.1 Qualification. Loudspeakers furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.2 Materials. Materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the loudspeaker to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.2.1 Aluminum. Aluminum shall conform to QQ-A-250/2.

3.3 Design and construction.

3.3.1 Dimensions. The dimensions of the loudspeaker shall be as shown in figure 1.

3.3.2 Width of air gap. The air gap for the voice coil in the permanent magnet shall be such that the difference between the minimum width of the radial air gap and the maximum thickness of the voice coil shall be at least 0.014 inch.

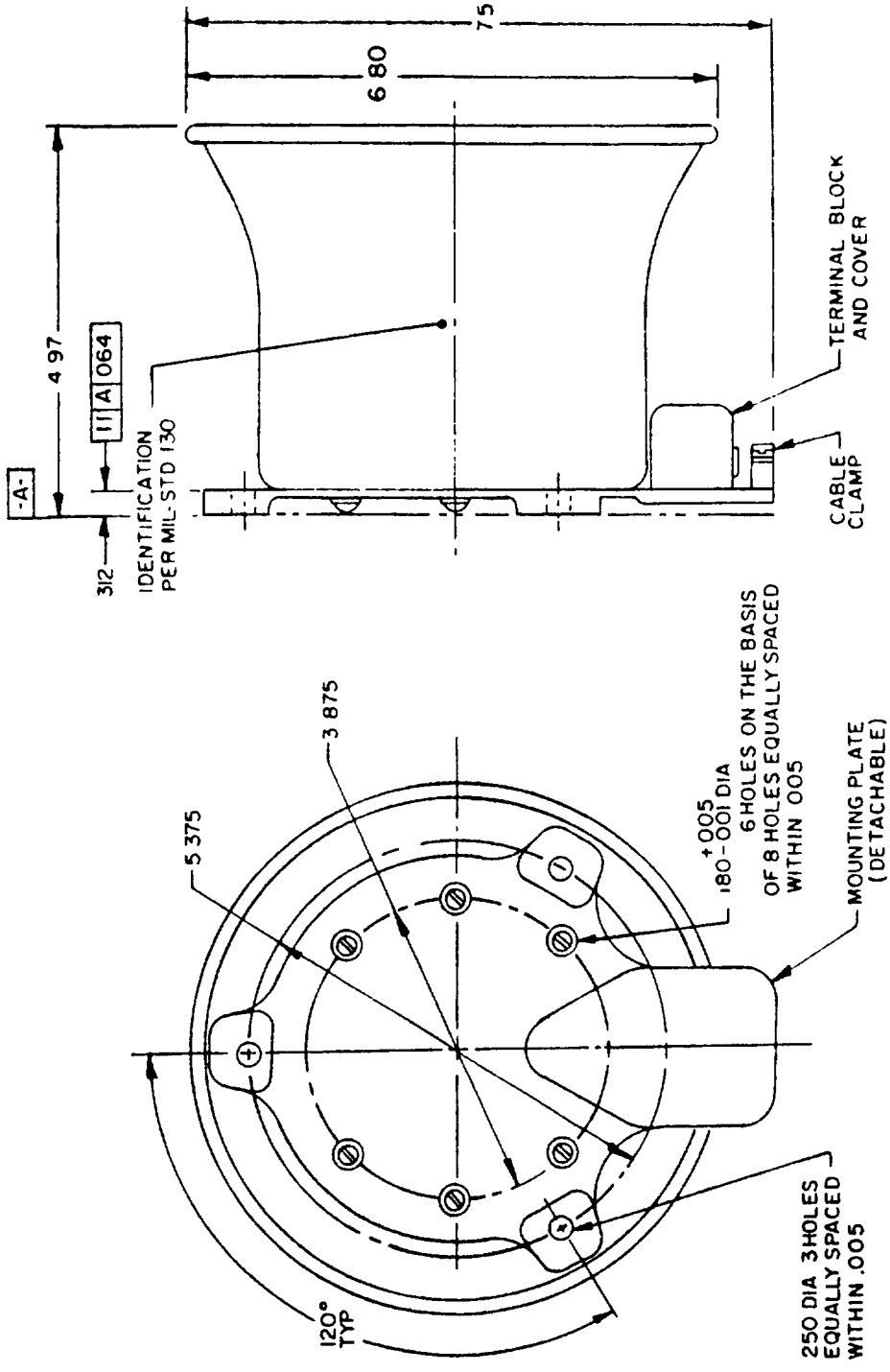
3.3.3 Weight. The weight of the loudspeaker shall not exceed 1.6 kilograms.

3.3.4 Color. The loudspeaker color shall be lusterless gray, color #36270 per FED-STD-595.

3.3.5 Terminal board. The voice-coil lead wire of the loudspeaker shall be brought to a terminal board on the mounting plate of the loudspeaker assembly, as shown on figure 1. The terminal board shall be covered with a metal protective shield with a grommet to permit the entry of a 5/16-inch diameter cable. The terminal board shall be constructed of a suitable high-grade insulating material and shall contain a minimum of two screws which will accept a No. 6 wire terminal.

3.4 Voice coil impedance. When loudspeakers are tested as specified in 4.7.2, the impedance of the voice coil, mounted in a complete loudspeaker assembly, shall be a minimum of 12.5 ohms.

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INCHES	MM	INCHES	MM
001	.03	3 875	98.43
005	.13	4 97	126.24
180	4 57	5 375	136 53
250	6 35	6 80	172 12
312	7 92	7 5	195 50

NOTES:
1 Dimensions are in inches
2 Unless otherwise specified tolerances are ± .015 (.38 mm) for three place decimals and ± .06 (1 52 mm) for two place decimals

FIGURE 1. Outline drawing loudspeaker LS 211/AIC-13

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3.5 Voice coil polarity (see 4 7.3). When tested as specified in 4 7 3 , the polarity of the voice coil shall be determined and the positive terminal of the voice coil shall be marked with a red dot The marking shall not interfere with the electrical connections and shall not be impaired by soldering or unsoldering operations.

3.6 Stray magnetic field. When tested as specified in 4.7 4, the loudspeaker shall not cause a deflection of more than 5° on a magnetic compass.

3.7 Sensitivity.

3.7.1 At sea level. When loudspeakers are tested as specified in 4 7.5 1, the average sensitivity above a reference level of 0.0002 dyne per square centimeter for each of the indicated frequency bands, shall be not less than the values shown in table I.

TABLE I Sensitivity.

Frequency band	Average sensitivity
Hz	dB
500 - 800	97 0
800 - 1250	102.0
1250 - 2000	103 0
2000 - 3200	106 0
3200 - 5000	103 5
500 - 5000	103.0

3.7.2 At altitude. When loudspeakers are tested as specified in 4 7 5 2, the sensitivity measured at an altitude of 15,000 feet, shall not differ by more than ±3 dB from the sensitivity measured at ground level

3.8 Frequency response When loudspeakers are tested as specified in 4.7.6, the frequency response shall be within the limits of figure 2, except that sharp peaks and dips may extend beyond these limits provided the width of the extension is not greater than 0.1 octave at the limit, and the number of extensions does not exceed 1 peak or dip per octave below 3,000 Hz, and 3 peaks or dips above 3,000 Hz.

3.9 Directional characteristics When loudspeakers are tested as specified in 4 7.7, the sound pressure measured 4 feet away from the loudspeaker in a free field at the indicated angles in all four quadrants with respect to the principal axis shall not deviate from the sound pressure measured on the principal axis by an amount greater than the values shown in table II for a frequency of 1,000 Hz The same input signal shall be applied for both on-axis and off-axis sound pressure measurements

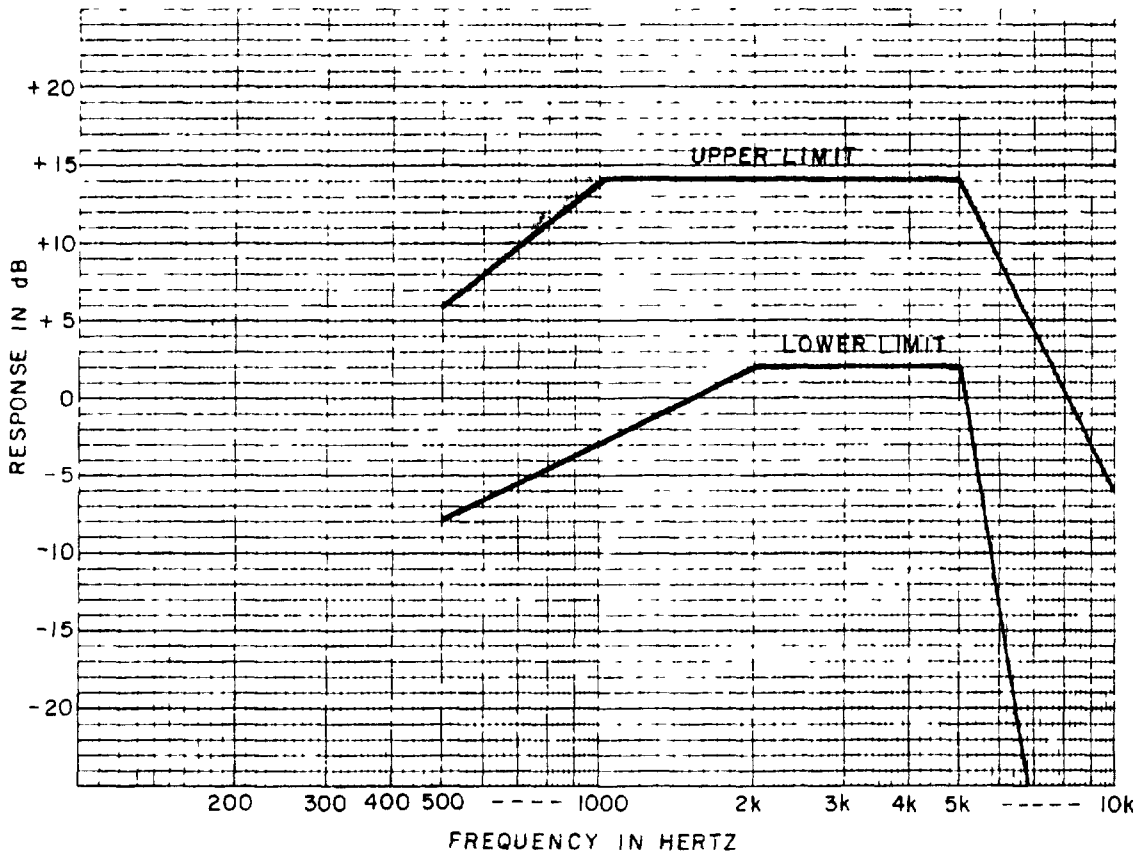
TABLE II. Directional characteristics

Frequency	Sound pressure deviation	
Hz	dB	
1,000	15°	30°
	-3.0	-6.0

3 10 Linearity When loudspeakers are tested as specified in 4 7 8, the acoustic output of the loudspeaker shall not deviate from a linear function of the input voltage by more than 1 dB.

3.11 Distortion. When loudspeakers are tested as specified in 4 7.9, the harmonic distortion in the acoustic output shall not exceed the values given in table III.

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0 = 95dB ± 2dB

FIGURE 2 Frequency response

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TABLE III. Distortion

Frequency range	Signal input	
	3.9 volts	8 8 volts
Hz	percent	percent
500 - 1,000	9	14
1,000 - 6,000	3 5	5

3 12 Acoustic quality. When loudspeakers are tested as specified in 4 7 10, there shall be no buzzes, rattles, or other spurious sounds that would impair the quality of reproduced audio signal in the output

3 13 Dielectric withstanding voltage When tested as specified in 4 7.11, there shall be no arcing or breakdown of the voice coil insulation Following the test, the loudspeaker shall meet the following requirements

Voice coil impedance variation - - - - - -	Not more than 1 ohm maximum from the initial measured value,

3.14 Life When tested as specified in 4.7.12, the loudspeaker shall meet the following requirements

Voice coil impedance variation - - - - - -	Not more than 1 ohm maximum from the initial measured value, except when measurements are made immediately after hot and cold exposures of the last cycle (see 3 4)
Sensitivity variation - - - - - - - - - - -	Not more than ±3 dB from the initial measured value (see 3 7.2).
Frequency response - - - - - - - - - - -	As specified in 3 8
Dielectric withstanding voltage - - - - -	As specified in 3 13
Visual examination - - - - - - - - - - -	There shall be no visual or mechanical failures (see table VIII)

3.15 Endurance When tested as specified in 4 7.13, loudspeakers shall meet the following requirements

Voice coil impedance variation - - - - - -	Not more than 1 ohm maximum from the initial measured value, except when measurements are made immediately after hot and cold exposures of the last cycle (see 3.4).
Sensitivity variation - - - - - - - - - -	Not more than ±2 dB from the initial measured value (see 3.7.2).
Frequency response- - - - - - - - - - - -	As specified in 3 8
Dielectric withstanding voltage - - - - -	As specified in 3.13
Visual examination - - - - - - - - - - - -	There shall be no visual or mechanical failures (see table VIII)

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3.16 Temperature cycling. When tested as specified in 4.7.14, the loudspeaker shall meet the following requirements

Voice coil impedance variation	Not more than 1 ohm maximum from the initial measured value, except when measurements are made immediately after hot and cold exposures of the last cycle (see 3.4).
Sensitivity variation	Not more than ±3 dB from the initial measured value (see 3.7.2)
Frequency response	As specified in 3.8
Dielectric withstanding voltage	As specified in 3.13
Visual examination	There shall be no visual or mechanical failures (see table VIII)

3.17 Vibration. When tested as specified in 4.7.15, the loudspeakers shall meet the following requirements

Voice coil impedance variation	Not more than 1 ohm maximum from the initial measured value, except when measurements are made immediately after hot and cold exposures of the last cycle (see 3.4)
Sensitivity variation	Not more than ±2 dB from the initial measured value (see 3.7.2)
Frequency response	As specified in 3.8
Dielectric withstanding voltage	As specified in 3.13

3.18 Shock. When tested as specified in 4.7.16, the loudspeakers shall meet the following requirements

Voice coil impedance variation	Not more than 1 ohm maximum from the initial measured value, except when measurements are made immediately after hot and cold exposures of the last cycle (see 3.4)
Sensitivity variation	Not more than ±2 dB from the initial measured value (see 3.7.2)
Frequency response	As specified in 3.8
Dielectric withstanding voltage	As specified in 3.13

3.19 Pressure cycling. When tested as specified in 4.7.17, the loudspeakers shall meet the following requirements

Voice coil impedance variation	Not more than 1 ohm maximum from the initial measured value, except when measurements are made immediately after hot and cold exposures of the last cycle (see 3.4)
Sensitivity variation	Not more than ±2 dB from the initial measured value (see 3.7.2)
Frequency response	As specified in 3.8
Dielectric withstanding voltage	As specified in 3.13

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3 20 Explosive decompression When tested as specified in 4.7 18, the loudspeakers shall meet the following requirements

Voice coil impedance variation - - - - - -	Not more than 1 ohm maximum from the initial measured value, except when measurements are made immediately after hot and cold exposures of the last cycle (see 3 4)
Sensitivity variation - - - - - - - - - -	Not more than ±2 dB from the initial measured value (see 3.7 2)
Frequency response - - - - - - - - - - - -	As specified in 3 8
Dielectric withstanding voltage - - - - -	As specified in 3 13
Visual examination - - - - - - - - - - - -	There shall be no visual or mechanical failures (see table VIII)

3 21 Humidity When tested as specified in 4 7 19, loudspeakers shall meet the following requirements

Voice coil impedance variation - - - - - -	Not more than 1 ohm maximum from the initial measured value, except when measurements are made immediately after hot and cold exposures of the last cycle (see 3 4)
Sensitivity variation - - - - - - - - - -	Not more than ±2 dB from the initial measured value (see 3 7 2)
Frequency response - - - - - - - - - - - -	As specified in 3 8
Dielectric withstanding voltage - - - - -	As specified in 3 13
Visual examination - - - - - - - - - - - -	There shall be no visual or mechanical failures (see table VIII)

3 22 Fungus When tested as specified in 4 7 20, loudspeaker shall meet the following requirements

Voice coil impedance variation - - - - - -	Not more than 1 ohm maximum from the initial measured value, except when measurements are made immediately after hot and cold exposures of the last cycle (see 3 4)
Sensitivity variation - - - - - - - - - -	Not more than ±2 dB from the initial measured value (see 3 7 2).
Frequency response - - - - - - - - - - - -	As specified in 3 8.
Dielectric withstanding voltage- - - - - -	As specified in 3.13
Visual examination - - - - - - - - - - - -	There shall be no visual or mechanical failures (see table VIII).

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3 23 Salt fog. When tested as specified in 4.7.21, loudspeakers shall meet the following requirements

Voice coil impedance variation - - - - - -	Not more than 1 ohm maximum from the initial measured value, except when measurements are made immediately after hot and cold exposures of the last cycle (see 3.4).
Sensitivity variation - - - - - - - - - -	Not more than ±2 dB from the initial measured value (see 3.7 2).
Frequency response - - - - - - - - - -	As specified in 3.8
Dielectric withstanding voltage - - - - -	As specified in 3.13
Visual examination - - - - - - - - - - - -	There shall be no visual or mechanical failures (see table VIII).

3 24 Marking Equipment assemblies and parts shall be marked for identification in accordance with MIL-STD-130

3 25 Workmanship The loudspeaker shall be processed in such a manner as to be in accordance with requirement 9 of MIL-STD-454

4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4.1 1 Test equipment and inspection facilities Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4 2 Classification of inspections The inspections specified herein are classified as follows

(a) Materials inspection (see 4 3)
(b) Qualification insepction (see 4 5)
(c) Quality conformance inspection (see 4 6)

4 3 Materials inspection Material inspection shall consist of certification supported by verifying data that aluminum used in fabricating the loudspeaker, is in accordance with the QQ-A-250/2, prior to such fabrication.

4.4 Inspection conditions Unless otherwise specified, loudspeakers shall be tested under the following conditions

Temperature - Room ambient, +15°C (+59°F) to +35°C (+95°F)
Pressure - Normal atmospheric
Humidity - Room ambient up to 90 percent relative humidity.

All acoustical testing shall be made a free field environment (see 6.4)

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4 5 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production

4.5.1 Sample size. Six loudspeakers shall be subjected to qualification inspection.

4.5.2 Inspection routine. The sample shall be subjected to the inspections specified in table IV, in the order shown. All sample units shall be subjected to the inspections of group I The sample shall then be divided equally into two groups, one shall be subjected to the group II inspections, and the other shall be subjected to group III inspections.

4.5.3 Failures. Failures in excess of those allowed in table IV shall be cause for refusal to grant qualification approval

TABLE IV. Qualification inspection.

Examination or test	Requirement paragraph	Method paragraph	Number of samples required	Number of failures permitted
Group I				
Visual and mechanical examination --	3 2, 3.3, 3 24 and 3.25	4.7.1	6	0
Voice coil impedance - - - - - - - -	3 4	4.7.2		
Voice coil polarity- - - - - - - - -	3 5	4 7.3		
Stray magnetic field - - - - - - - -	3.6	4 7 4		
Sensitivity- - - - - - - - - - - - -	3.7	4.7.5		
Frequency response - - - - - - - - -	3 8	4.7.6		
Directional characteristics- - - - -	3 9	4.7.7		
Linearity - - - - - - - - - - - - -	3.10	4 7 8		
Distortion- - - - - - - - - - - - -	3.11	4 7.9		
Acoustic quality - - - - - - - - - -	3.12	4 7 10		
Dielectric withstanding voltage- - -	3 13	4 7.11		
Group II				
Life - - - - - - - - - - - - - - - -	3.14	4.7.12	3	0
Endurance- - - - - - - - - - - - - -	3.15	4.7.13		
Temperature cycling- - - - - - - - -	3.16	4 7.14		
Vibration- - - - - - - - - - - - - -	3.17	4.7.15		
Shock- - - - - - - - - - - - - - - -	3 18	4.7 16		
Group III				
Pressure cycling - - - - - - - - - -	3 19	4.7.17	3	0
Explosive decompression- - - - - - -	3.20	4 7.18		
Humidity- - - - - - - - - - - - - -	3.21	4.7.19		
Fungus- - - - - - - - - - - - - - -	3.22	4.7 20		
Salt fog- - - - - - - - - - - - - -	3.23	4.7.21		

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4.5.4 Retention of qualification. To retain qualification, the supplier shall forward a report at 6-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

(a) A summary of the results of the tests performed for inspection of product for delivery, group A, indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests to all reworked lots shall be identified and accounted for.

(b) The results of tests performed for qualification verification inspection, group C, including the number and mode of failures. The test report shall include results of all qualification verification inspection tests performed and completed during the 12-month period. If the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 6-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the supplier shall immediately notify the qualifying activity at any time during the 6-month period that the inspection data indicates failure of the qualified product to meet the requirements of this inspection.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during 2 consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit the loudspeakers to testing in accordance with the qualification inspection requirements.

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections.

4.6.1.1 Inspection lot. An inspection lot shall consist of all loudspeakers produced under essentially the same conditions, and offered for inspection at one time.

4.6.1.2 Group A inspection. Group A inspection shall consist of the examination and tests specified in table V in the order shown.

4.6.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in table V. Major and minor defects shall be as defined in MIL-STD-105. The classification of defects for the visual and mechanical examination is specified in table VIII.

TABLE V. Group A inspection.

Examination or test	Requirement paragraph	Method paragraph	AQL (percent defective)	
			Major	Minor
Visual and mechanical examination	3.2, 3.3, 3.24 and 3.25	4.7.1	1.0	4.0
Voice coil impedance	3.4	4.7.2	1.0	None
Voice coil polarity	3.5	4.7.3	1.0	
Acoustic quality	3.12	4.7.10	1.0	None

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4.6.1.2 2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.6.1.3 Group B inspection Group B inspection shall consist of the tests specified in table VI and shall be made on sample units which have been subjected to and have passed group A inspection.

4.6 1.3.1 Sampling plan The sampling plan shall be in accordance with MIL-STD-105 for special inspection level S-4 The AQL shall be 1 percent defective

4.6 1.3 2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots

TABLE VI Group B inspection

Test	Requirement paragraph	Method paragraph
Sensitivity	3 7	4 7 5
Frequency response	3 8	4 7 6
Directional characteristics	3 9	4 7 7
Distortion	3 11	4 7 9
Dielectric withstanding voltage	3 13	4 7 11

4.6.1 3.3 Disposition of sample units Sample units which have passed all the group B inspection may be delivered on the contract or purchase order, if the lot is accepted.

4.6.2 Qualification verification inspection Qualification verification inspection shall consist of group C. Except where the results of these inspections show non-compliance with the applicable requirements (4 6.2 1 4), delivery of products which have passed groups A and B shall not be delayed pending the results of these inspections.

4.6.2 1 Group C inspection Group C inspection shall consist of the tests specified in table VII in the order shown Group C inspection shall be made on sample units selected from inspection lots which have passed groups A and B inspections

4 6 2 1 1 Sampling plan Group C inspection shall be performed once each 6 months on six sample units selected without regard to their quality from units produced during the period The sample shall be divided equally into two groups and subjected to the tests of subgroups 1 and 2 of table VII

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TABLE VII Group C inspection

Test	Requirement paragraph	Method paragraph
Subgroup 1 (3 sample units)		
Stray magnetic field	3 6	4 7.4
Endurance	3 15	4 7.13
Temperature cycling	3 16	4 7.14
Vibration	3.17	4 7 15
Shock	3 18	4 7 16
Subgroup 2 (3 sample units)		
Life	3.14	4 7 12
Linearity	3 10	4 7 8
Pressure cycling	3 19	4 7 17
Explosive decompression	3 20	4 7 18
Humidity	3 21	4 7 19
Fungus	3 22	4 7 20
Salt fog	3 23	4 7 21

4 6 2 1 2 Failures If one or more sample units fail to pass group C inspection, the sample shall be considered to have failed

4 6 2 1 3 Disposition of sample units Sample units which have been subjected to group C inspection shall not be delivered on the contract or purchase order

4 6.2 1 4 Noncompliance If a sample fails to pass group C inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc , and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government) Groups A and B inspections may be reinstituted, however, final acceptance shall be withheld until the group C reinspection has shown that the corrective action was successful In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the contracting officer

4.6.3 Inspection of preparation for delivery Except for commercial packaging, the sampling and inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance requirements of MIL-P-116 The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129 The inspection of commercial packaging shall be as specified in the contract or purchase order (see 6.2)

4 7 Methods of examination and test

4 7 1 Visual and mechanical examination The loudspeaker shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.2, 3 3, 3.24, and 3 25). Unless otherwise specified (see 6 2), the defects shall be classified as specified in table VIII

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4.7.2 Voice coil impedance (see 3.4). The loudspeaker shall be hand held on the side opposite the cone. There shall be no obstruction within 3 feet of the front of the cone. The voice coil terminals shall be connected in series with a variable resistor across the output of a low impedance 1,000 Hz signal source. The voltage drops across the voice coil and the variable resistor shall be determined with an electronic voltmeter. The value of the variable resistor shall be adjusted until the voltage drops are equal and the resistance of the variable resistor shall then be measured with a resistance bridge or other suitable equipment. The voice coil impedance shall be considered as numerically equal to the measured value of resistance of the variable resistor.

4.7.3 Voice coil polarity (see 3.5). A direct current voltage of 4.5 ± 1.5 volts shall be applied to the terminals of the voice coil so that it causes the diaphram of the loudspeaker to move in a direction away from the magnetic assembly. The terminal connected to the positive voltage potential shall be marked as the positive terminal of the voice coil.

4.7.4 Stray magnetic field (see 3 6) The stray magnetic field of the loudspeaker shall be determined as follows, by using a compass which meets the requirements of GG-C-560a Type II or engineering approved equivalent. The loudspeaker shall be placed with the geometric center of the magnet structure 20 inches ± 1 inch from the pivot point of the compass needle and in the plane of rotation of the needle (see figure 3). The loudspeaker shall be oriented in all directions and the maximum deflection of the compass observed. The test shall be made in a location substantially free from stray magnetic disturbances. The maximum deflection of the compass shall be determined as specified in 4.7.3.1 or 4.7.4.2 at the option of the manufacturer.

4.7.4 1 Constant compass location (six permanently mounted compasses.) With a compass permanently mounted in each location indicated in figure 3, the readings shall be observed on each compass as the loudspeaker is oriented in all directions.

4.7.4.2 Changing compass location (one movable compass). In each direction that the loudspeaker is oriented, the compass shall be relocated to each of the six compass locations and oriented so that the degree markings on the compass line up (within plus or minus .5 degrees) with the degree markings on the table as shown in figure 3. The reading on the compass shall be observed in each location for each direction the loudspeaker is oriented.

4.7.5 Sensitivity.

4.7.5.1 At sea level (see 3.7 1). A sensitivity measurement shall be made with a constant signal input 3.94 ± .01 volts in a free field at a point on the principal axis 4 feet ± 1 inch from the front of the loudspeaker. The average sensitivity for each frequency band may be obtained by measuring the area between the frequency response and any reference line over the appropriate frequency band, provided this line was established from a basic reference level of 0.0002 dyne per square centimeter

4.7.5.2 At altitude (see 3.7.2). The sensitivity test specified in 4.7 5.1 shall be repeated at a simulated altitude of 15,000 feet above sea level.

4.7.6 Frequency response (see 3.8) The loudspeaker shall be mounted and placed 4 feet ± 1 inch from a calibrated Western Electric Type 640AA (or equal) condenser microphone and amplifier. The microphone shall be placed on the axis of the loudspeaker. A constant voltage having a rms value of 3.94 ±.05 volts shall be applied to the voice coil terminals and the frequency varied continuously from 200 to 7,000 Hz The acoustic output shall be recorded on a direct-writing strip graph chart using an automatic plotter or curve tracer with a minimum writing speed of 10 inches per second and a maximum chart speed of 30 inches per minute, or, at the option of the manufacturer, point-to-point measurements may be made every 50 Hz from 200 to 600 Hz; every 100 Hz from 600 to 1,500 Hz, every 250 Hz from 1,500 Hz to 2,000 Hz, every 100 Hz from 2,000 to 2,500 Hz, every 250 Hz from 2,500 to 3,000 Hz, and every 500 Hz from 3,000 to 7,000 Hz with the data recorded and a graph drawn.

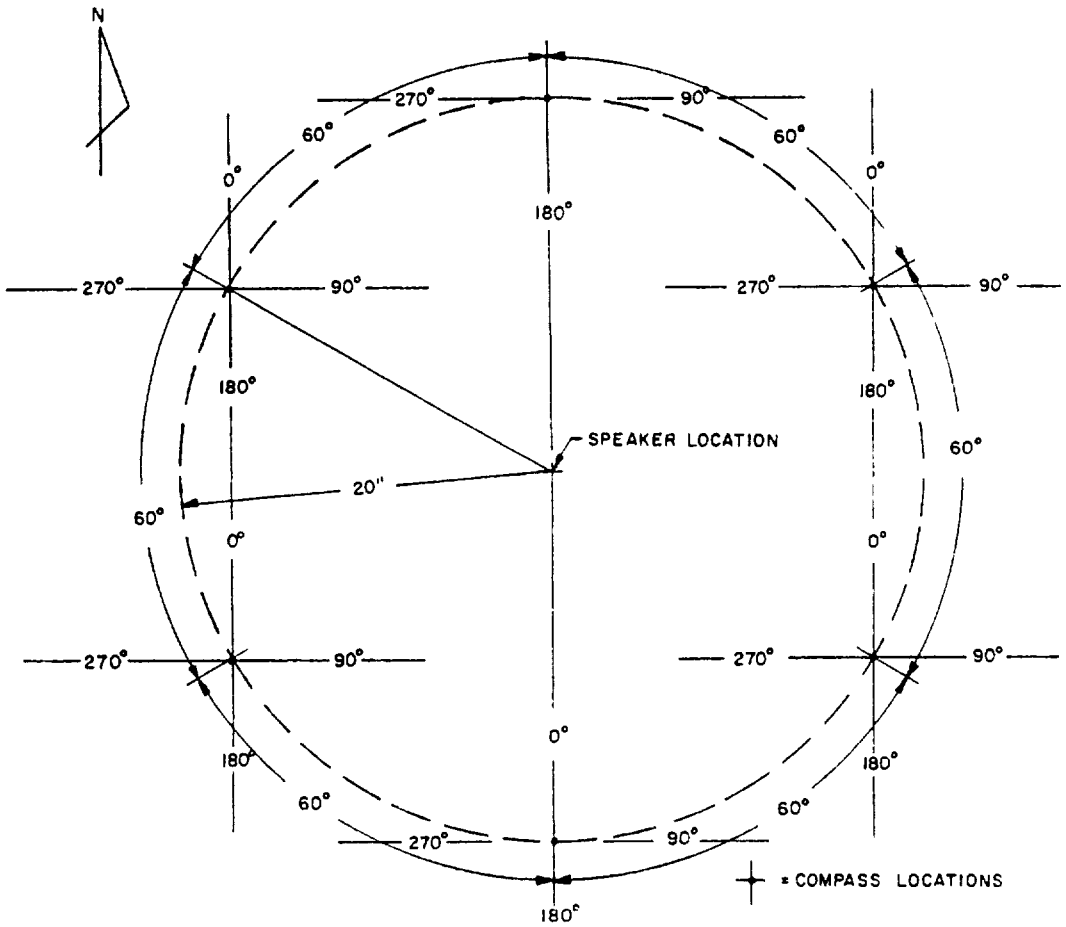
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TABLE VIII Classification of defects for visual and mechanical examination

Defect type	Classification	
	Major	Minor
Dimensions	Dimensions not as specified	---
Materials and finish	Materials not as specified Wrong or incomplete finish Large amounts of flaking, peeling or chipping of finish	Scratches cuts abrasions etc causing exposure of base metal or relatively small amounts of flaking peeling, or chipping
Parts	Missing parts Inoperative, improperly assembled or defective parts which could cause the loudspeaker to fail in service Wrong parts	Defective parts which would reduce efficiency of use but not cause failure in service Cracks or chipped surfaces having no effect on the functioning, assembly maintenance, or life of the loudspeaker
Marking	Marking missing, illegible, or incorrect	Markings dirty or smudged, but legible
Foreign objects	Any metallic foreign object, not firmly attached [1], which could cause a short circuit or acoustical malfunctioning of the loudspeaker. Any nonmetallic foreign object such as insulation, dirt, or phenolic chips which could cause acoustical malfunctioning of the loudspeaker.	Any metallic or nonmetallic foreign object which affects appearance but which could not cause acoustical malfunctioning of the loudspeaker
Soldering	Improper wrap - Less than 1/2 turn Unsoldered joint - Solder not applied where intended Insufficient solder - Minimum dimension of solder bridge less than twice the diameter of the wire or less than 3/32 inch, whichever is greater Entire area of contact between wire and terminal not joined by solder bridge Cold solder joint - Chalky appearance, lacks metallic luster, presents rough "pile-up" appearance, movement of wire or solder upon pick application Rosin joint - Presence of excess rosin, relative movement of wire or solder upon pick application Insulation in terminal hole - Solder over insulation no appearance of visible wire contour	Improper wrap - 1/2 turn or more but less than one turn Excess solder - Build-up solder or joint greater than necessary for good soldering usually resulting in obliteration of wire contour Cold solder joint - Chalky appearance lacks metallic luster, presents rough "pile-up" appearance no relative action between wire and solder upon pick application
Wiring	Broken strands - More than 20 percent except in a 7-strand conductor, more than 2 broken strands Insulation burned, abrased, pinched or deteriorated between two or more conductors, resulting in a potential short circuit Taut wire - Wire exhibits no slack and subsequent breakage may occur due to stress on terminal or part Insulation frayed to the extent that a potential short circuit exists	Broken strands - 20 percent or less In a 7-strand conductor 2 broken strands Insulation burned abrased pinched or deteriorated with exposure of bare wire but short circuit not possible Taut wire - Slight stress on conductor but not possibility of subsequent breakage

[1] Foreign objects that cannot be dislodged by the moderate application of pressure with a pick or spudger shall be considered to be firmly attached

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NOTE:

FIGURE 3. Stray magnetic field test table.

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4 7.7 Directional characteristics (see 3 9) The on-axis sound pressure shall be measured in a free field at a point 4 feet from the front of the loudspeaker on the principal axis. The off-axis sound pressure shall be measured in a free field at a point 4 feet ±1 inch from the front of the loudspeaker at angles of 15 degrees and 30 degrees with respect to the principal axis The vertices of the angles shall be the intersection point of the loudspeaker principal axis and the plane containing the front face of the loudspeaker. The same input signal, at 1,000 Hz shall be applied for both on-axis and off-axis sound pressure measurements

4 7 8 Linearity (see 3.10) The acoustic output of the loudspeaker in dB relative to 0 0002 dyne per square centimeter shall be measured by using a warble signal having a frequency range from 750 Hz to 1,250 Hz and a rate of sweep of 5-1/2 times per second Input voltages shall be 0 313, 0.557, 0 99, 1 76, 3 13, 5.57, 9 9, and 17 6 volts The acoustic output shall be plotted as a function of the input voltage

4.7 9 Distortion (see 3.11) The harmonic distortion in the acoustic output of the loudspeaker at a certain frequency of the fundamental shall be determined by the following formula

$$\text{RMS distortion (in percent)} = 100 \frac{\sqrt{P_2^2 + P_3^2 + \text{-------} Pn^2}}{\sqrt{P_1^2 + P_2^2 + P_3^2 + \text{--} Pn^2}}$$

Where P_1 is the pressure amplitude of the fundamental, and P_2, P_3, P_4, etc , are the pressure amplitudes of the harmonic components in the output, the distortion may also be determined by using a total distortion analyzer

4 7 10 Acoustic quality (see 3.12) A constant voltage having an rms value of 4 volts and a frequency between 500 and 5,000 Hz shall be applied to the input terminals of the loudspeaker, and the frequency shall be varied from 500 Hz to 5,000 Hz and back to 500 Hz in not less than 1 minute

4 7 11 Dielectric withstanding voltage (see 3 13) The loudspeaker shall be tested in accordance with method 301 of MIL-STD-202 The following details shall apply

(a) Test voltage - 400 volts
(b) Nature of potential - 60 Hz, ac
(c) Points of application - The test voltage shall be applied between one of the voice coil terminals and the loudspeaker frame, simultaneously, the voice coil shall be excited with a minimum of 2 85 volts root mean square (rms) at 300 Hz
(d) Measurement after test - Voice coil impedance shall be measured as specified in 4.7.2

4 7.12 Life (see 3.14) The loudspeaker shall be placed within a test chamber wherein a temperature of +71°C (+160°F) is maintained. The loudspeaker shall be placed in the test chambers with the front face exposed The loudspeaker shall be operated for 3 hours with a signal input of 17 6 volts and a frequency of 1,200 Hz applied to the voice coil in alternate 15 second "on" and 15 seconds "off" periods At the conclusion it shall be immediately transferred to a chamber having an internal temperature of -55°C (-67°F) exposed to this temperature for 3 hours, and allowed to stablize at that temperature The loudspeaker shall be operated for 3 hours with a signal input of 17 6 volts and a frequency of 1,200 Hz applied to the voice coil in alternate 15 seconds "on" and 15 seconds "off" periods. Following the test, the loudspeaker shall be exposed to room ambient conditions and allowed to stabilize at that temperature At the conclusion it shall be transferred to a chamber having an internal

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temperature of 25°C (+77°F), this temperature for hours. The loudspeaker shall be operated for 100 hours with a signal input of 17.6 volts and a frequency of 1,200 Hz applied to the voice coil in alternate 15 seconds "on" and 15 seconds "off" periods. Following the test, the loudspeaker shall be allowed to stabilize at room ambient temperature and then operated for 240 hours at a continuous value of 3.94 volts rms. Following the test, voice coil impedance, sensitivity, frequency response and dielectric withstanding voltage shall be measured in accordance with 4.7.2, 4.7.5.2, 4.7.6, and 4.7.11, respectively, and the loudspeaker shall be examined for evidence or damage.

4.7.13 Endurance (see 3.15). The loudspeaker shall be operated for 150 hours + 0 or -2 hrs at a simulated altitude of 15,000 feet with a warble signal input of 17.6 volts, having a frequency range from 1,000 Hz to 3,000 Hz and a rate of sweep of approximately six times per second. The signal shall be applied intermittently 1 minute ±.5 seconds in every 3 minutes. Following the test, voice coil impedance, sensitivity, frequency response, and dielectric withstanding voltage shall be measured in accordance with 4.7.2, 4.7.5.2, 4 7.6, and 4.7.11, respectively, and the loudspeaker shall be examined for evidence of damage.

4.7 14 Temperature cycling (see 3.16) The loudspeaker shall be placed in the test chamber with the front face exposed The loudspeaker shall be exposed to a temperature of +85°C + 3° or -0° (+185°F) for 1 hour, placed in room temperature for 15 minutes, then exposed to a temperature of -65°C (-85°F) for 1 hour ±1 minute, and again placed in room temperature for 15 minutes ±1 minute The loudspeaker shall be subjected to 10 such temperatures cycles Immediately after the +85°C exposure and -65°C exposure during the last cycle, and 1 hour after completion of the test, voice coil impedance, sensitivity, frequency response, and dielectric withstanding voltage shall be measured in accordance with 4.7.2, 4 7.5, 4 7.6, and 4.7.11, respectively, and the loudspeaker shall be examined for evidence of damage.

4.7 15 Vibration (see 3.17) The loudspeaker shall be tested in accordance with method 514 of MIL-STD-810. The following details and exceptions shall apply

(a) Procedure - 1, part 1
(b) Test level curve - At figure 514.1-1
(c) Category (a) equipment
(d) Measurements after the test - Within 1 hour after the test, voice coil impedance, sensitivity, frequency response, and dielectric withstanding voltage shall be measured in accordance with 4 7.2, 4.7.5 2, 4.7.6, and 4.7.11, respectively, and the loudspeaker shall be examined for evidence of damage

4.7 16 Shock (see 3.18) The loudspeaker shall be tested in accordance with test method 516 of MIL-STD-810 The following details and exceptions shall apply

(a) Procedure I.
(b) Shock pulse.
  (1) Sawtooth wave shape
  (2) 20g peak value.
  (3) 11 ms duration.
(c) Measurements after the test. Within 1 hour after the test, voice coil impedance, sensitivity, frequency response, and dielectric withstanding voltage shall be measured in accordance with 4.7.2, 4.7.5.2, 4.7.6, and 4.7 11, respectively, and the loudspeaker shall be examined for evidence of damage.

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4 7 17 Pressure cycling (see 3 19) The loudspeaker shall be placed in the test chamber with the face of the loudspeaker exposed and the absolute internal pressure of the chamber reduced to 3.44 inches of mercury (corresponding to an altitude of 50,000 feet above sea level) and allowed to remain at the reduced pressure for 5 minutes ± 5 seconds The pressure shall be increased again to that prevailing at ground level and the loudspeaker shall be allowed to remain at atmospheric pressure for 5 minutes ± 5 seconds The rate of change of pressure shall correspond to approximately 50,000 feet per minute The loudspeaker shall be subjected to three such pressure cycles and shall be operated throughout the test with a warble signal input of 1 6 volts, having a frequency range from 1,000 Hz to 3,000 Hz, and a rate of sweep of approximately six times per second Following the test, voice coil impedance, sensitivity, frequency response, and dielectric withstanding voltage shall be measured in accordance with 4 7 2, 4.7 5 2, 4.7 6, and 4 7 11, respectively, and the loudspeaker shall be examined for evidence of damage

4 7 18 Explosive decompression (see 3 20) The loudspeaker shall be subjected to 10 consecutive explosive decompressions from an altitude of 8,000 feet to an altitude of 35,000 feet (corresponding to pressures of 22 2 inches of mercury to 7 01 inches of mercury) Each complete decompression or change of pressure shall take place in 0 1 second or less The loudspeaker shall be placed in the test chamber with the front face of the loudspeaker exposed Following the test, voice coil impedance, sensitivity, frequency response, and dielectric withstanding voltage in accordance with 4 7 2, 4 7 5 2, 4 7.6, and 4.7 11, respectively, and the loudspeaker shall be examined for evidence of damage

4 7 19 Humidity (see 3.21) The loudspeaker shall be subjected to humidity testing in accordance with method 507 of MIL-STD-810 The following details and exceptions shall apply

(a) Procedure I.
(b) Measurements after the test Within 1 hour after the test, voice coil impedance, sensitivity, frequency response, and dielectric withstanding voltage, shall be measured in accordance with 4 7 2, 4 7 5 2, 4 7 6, and 4 7 11, respectively, and the loudspeaker shall be examined for evidence of damage

4 7 20 Fungus (see 3 22) The loudspeaker shall be tested in accordance with method 508 of MIL-STD-810 The following details shall apply

(a) Measurements after the test Within 1 hour after the test, voice coil impedance, sensitivity, frequency response, and dielectric withstanding voltage, shall be measured in accordance with 4 7 2, 4 7 5 2, 4 7 6, and 4 7 11, respectively, and the loudspeaker shall be examined for evidence of damage

4 7 21 Salt fog (see 3 23) The loudspeaker shall be tested in accordance with method 509 of MIL-STD-810 The following detail shall apply

(a) Measurements after the test Within 1 hour after the test, voice coil impedance, sensitivity, frequency response, and dielectric withstanding voltage shall be measured in accordance with 4 7 2, 4 7.5 2, 4.7 6, and 4 7 11, respectively, and the loudspeaker shall be examined for evidence of damage

5 PREPARATION FOR DELIVERY

5.1 Preservation-packaging Preservation-packaging shall be level A or C, or as specified (see 6 2)

5 1 1 Level A

5 1 1 1 Cleaning Loudspeakers shall be cleaned in accordance with MIL-P-116, process C-1

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5 1 1 2 Drying. Loudspeakers shall be dried in accordance with MIL-P-116

5 1 1 3 Preservative application Preservatives shall not be used

5 1 1 4 Unit packaging Loudspeakers shall be individually packaged in accordance with MIL-P-116, submethod IC-2 insuring compliance with the applicable requirements of that specification

5 1 1 5 Intermediate packaging Not required.

5 1 2 Level C. Loudspeakers shall be clean, dry and packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity This level may conform to the supplier's commercial practice when such meets the requirements of this level

5 2 Packing Packing shall be level A, B, or C, or as specified (see 6 2)

5 2 1 Level A The packaged loudspeakers shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements In lieu of the closure and waterproofing requirement in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing allseams, corners, and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76 Banding (reinforcement requirements) shall be applied in accordance with the appendix PPP-B-636 using nonmetallic or tape banding only

5 2 2 Level B The packaged loudspeakers shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements Closures shall be in accordance with the appendix thereto

5 2 3 Level C The packaged loudspeakers shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity These packs shall conform to the applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level

5 2 4 Unitized loads Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more Quantities less than 40 cubic feet need not be unitized Unitized loads shall be uniform in size and quantities to the greatest extent practicable

5.2 4 1 Level A Loudspeakers, packed as specified in 5 2 1, shall be unitized on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

5.2 4 2 Level B. Loudspeakers, packed as specified in 5 2 2, shall be unitized as specified in 5 2 4 1 except that the fiberboard caps shall be class domestic

5 2.4 3 Level C Loudspeakers, packed as specified in 5 2 3, shall be unitized with pallets and caps of the type, size and kind commonly used for the purpose. These unitized loads shall conform to the applicable carrier rules and regulations.

5.3 Marking In addition to any special marking required by the contract or purchase order (see 6.2), each unit package, exterior container and unitized load shall be marked in accordance with MIL-SID-129 The loudspeaker shall be identified as loudspeaker, magnetic LS-211/AIC-13

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5 4 General.

5 4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistant with the protection required and shall contain equal quantitities of identical stock numbered items to the greatest extent practicable

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4 6.3.

5 4 3 Army procurements.

5 4.3.1 Level A (maximum military protection) and level B (minimum military protection) packing For level A packing when quantities per destination are less than a unitized load, the fiberboard containers shall not be banded but shall be placed in a close fitting box conforming to PPP-B-601, overseas type, PPP-B-621, class 2 style 4 or PPP-B-585, class 3, style 2 or 3 Closure and strapping shall be in accordance with applicable container specification except tnat metal strapping shall conform to QQ-S-781, type I, finish B When the gross weight exceeds 200 pounds or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification If not described in the container specification, the skids shall be applied in a manner which will adequately support the item and facilitate the use of material handling equipment For level B packing, fiberboard boxes shall be weather resistant as specified in level A and the containers shall be banded (see 5 2 1 and 5 2 2)

5 4.3 2 Level A and B unitization For level A and B unitization, the fiberboard caps shall be weather resistant and softwood pallets conforming to NN-P-71, type IV, size 2 shall be used. The loads shall be bonded to the pallets by strapping conforming to QQ-S-781, type I, finish A or shrink film (see 5 2 4 1 and 5 2.4.2)

5 4.3.3 Commercial unit packaging and packing Commercial unit packaging and packing shall conform to the requirements of 5 1 2 and 5 2 3 as applicable

5 4 3 4 Commercial marking All unit and exterior containers shall, as a minimum, be marked with the following noun nomenclature, national stock number (or part number when the NSN is not given), Government contract or purchase order number, quantity, contractor's name and any additional marking which may be required by the contract or the contractor's policy or procedures Exterior containers shall also be marked with the appropriate address All markings shall be applied by any means providing legibility

6. NOTES

6.1 Intended use. Loudspeaker, permanent magnetic LS-211/AIC-13 is intended to be used as part of public address set AN/AIC-13 to transmit audio messages at low altitudes or at ground level within a military aircraft or from a military aircraft on the ground to personnel in the immediate vicinity This loudspeaker is a horn type unit with a frequency response of 200-7000 Hz, nominal 16 ohm voice coil impedance, and a 20-watt peak power output.

6.2 Ordering data Procurement documents should specify the following

(a) Title, number, and date of this specification.
(b) Inspection of commercial packaging (see 4.6 3).
(c) Applicable document for classification of visual anD mechanical defects if other than table VIII (see 4 7.1)
(d) Levels of preservation-packaging and packing required (see 5.1 and 5 2)
(e) Special marking, if required (see 5.3)

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6.3 Qualification With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable qualified products list whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Air Force Logistics Command, Wright-Patterson Air Force Base, Dayton, Ohio 45433. Information pertaining to qualification of products may be obtained from AFLC/MMGE, Wright-Patterson A.F.B. for administration of the qualified products list.

6.4 Free field environment. The acoustic environment shall simulate free-field conditions to the extent that the inverse-pressure versus distance law should hold within plus or minus 1 dB at all frequencies at which measurements are made Ambient noise should not affect the measurements to an extent greater than plus or minus 1 dB

6.5 Design activity The design agency for this item is AFLC/MMGE.

6.6 Changes from previous issue Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes

Custodians
Army - EL
Navy - EC
Air Force - 85

Review activities
Army - EL
Navy - EC
Air Force - 80
DSA - ES

User activities
Army - ME, AT, AV, MU
Navy - OS
Air Force - 80, 85

Preparing activity
Air Force - 85

(Project 5965-0142)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R25[illegible]

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-19978/16 Amnd 2 Capacitors, Fixed, Plastic (Or Paper-Plastic) Dielectric

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRAC

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426

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DOC IX

MIL-L-25547B
AMENDMENT 1
5 May 1975

MILITARY SPECIFICATION

LOUDSPEAKER, PERMANENT MAGNET LS-211/AIC-13

This amendment forms a part of Military Specification MIL-L-25547B, 6 February 1975, and is approved for use by all Departments and Agencies of the Department of Defense.

Page 17

4.7.12, line 13, after "temperature of 25°C (+77°F)," delete "this temperature for hours." and substitute "exposed to this temperature for 3 hours, and allowed to stabilize at that temperature."

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activity:
DSA - ES

Preparing activity:
Air Force - 85

(Project 5965-0152)

FSC 5965

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